prepare_spatial_variance_waffle_data.R

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Contents

`prepare_spatial_variance_waffle_data`
Prepare Spatial Variance Waffle Data

Description

Creates tile positions and precomputed trivariate fill colours for unit-level waffle plots.

Usage

```
prepare_spatial_variance_waffle_data(
  data_plot,
  vec_component_colours,
  n_waffle_rows = 5L,
  vec_required_components = base::c("Abiotic", "Spatial", "Associations"),
  ranking_column = "component_total_percentage",
  anchor_component = "Associations",
  method = base::c("HCL", "perc_avg")
)
```

Arguments

`data_plot` Data frame returned by [prepare_spatial_variance_plot_data()] and containing variance components for each observation.

`vec_component_colours` Named character vector mapping canonical component IDs to base HEX colours.

`n_waffle_rows` Positive integer giving the number of rows in each waffle panel.

`vec_required_components` Character vector of required canonical component IDs used for trivariate colour mixing.

`ranking_column` Single character string naming the column used for within-panel tile ordering. Defaults to ‘component_total_percentage‘.

anchor_component
: Optional single component ID used to select one row per observation after colour mixing. Defaults to ‘"Associations"‘ to preserve historical behaviour. When ‘NULL‘, the first value in ‘vec_required_components‘ is used.

method
: Character string selecting the colour-mixing method passed to [compute_variance_component_colour

Value

A tibble with one row per observation and columns ‘tile_col‘, ‘tile_row‘, ‘tile_fill_colour‘, and ‘point_colour‘.

Examples

```
data_plot <-
  tibble::tibble(
    scale = base::rep("local", 6),
    scale_id = base::c(
      "local_01",
      "local_01",
      "local_01",
      "local_02",
      "local_02",
      "local_02"
    ),
    resolution_id = base::rep("genus", 6),
    resolution_label = base::rep("Genus", 6),
    component = base::c(
      "Abiotic",
      "Spatial",
      "Associations",
      "Abiotic",
      "Spatial",
      "Associations"
    ),
    continent_id = base::c(
      "europe",
      "europe",
      "europe",
      "asia",
      "asia",
      "asia"
    ),
    component_total_percentage = base::c(50, 30, 20, 30, 30, 40)
  )

prepare_spatial_variance_waffle_data(
  data_plot = data_plot,
  vec_component_colours = base::c(
    "Abiotic" = "#D95F02",
    "Spatial" = "#7570B3",
    "Associations" = "#1B9E77"
  )
)
```

Index